



2025–2026 Undergraduate Catalog

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# Physics with Astrophysics Concentration (B.S.)



**Location(s):** [Dahlonega](#)

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120 hours

## Core IMPACTS – 42 hours

Refer to [Core IMPACTS](#)

**Institution:** No exceptions

**Mathematics:** MATH 1113 or MATH 1450

**Citizenship:** No exceptions

**Humanities:** No exceptions

**Writing:** No exceptions

**STEM:** CHEM 1211/1211L and CHEM 1212/1212L and MATH 1450 or MATH 2460 (if MATH 1450 was taken in IMPACTS Mathematics) recommended

**Social Sciences:** No exceptions

## Field of Study – 18 hours

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### Required Courses

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- PHYS 2211 – Principles of Physics I
- PHYS 2211L – Principles of Physics I Lab
- (or PHYS 2211K – Principles of Physics I )
- PHYS 2212 – Principles of Physics II
- PHYS 2212L – Principles of Physics II Lab
- (or PHYS 2212K – Principles of Physics II )

Select one:

- CSCI 1301 – Computer Science I
- CSCI 1301K – Computer Science I
- CSCI 1371 – Computing for Scientists and Engineers

### For Students with MATH 1113 in Mathematics

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- MATH 2460 – Calculus II
- MATH 2470 – Calculus III

### For Students with MATH 1450 in Mathematics

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- MATH 2470 – Calculus III
- 2 carryover hours from Core IMPACTS Mathematics and STEM
- 1 elective hour in any 1–2000 level ENGR or Core IMPACTS STEM course (except MATH 1113 or CSCI 1250) if CSCI 1371 is taken

## Major Requirements – 60 hours

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### Required Courses - 25 or 37 hours\*

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- PHYS 1000 – Physics and Society
- ASTR 3150 – Introductory Astrophysics
- PHYS 3111 – Mechanics I
- PHYS 3310 – Modern Physics

- PHYS 3310L – Modern Physics Laboratory
- PHYS 3411 – Electricity and Magnetism I
- PHYS 3610 – Thermal Physics
- PHYS 3991 – Research Practicum I: Planning and Proposing Research
- PHYS 4350 – Quantum Mechanics
- PHYS 4991 – Research Practicum II: Performing and Writing Scientific Research
- PHYS 4992 – Research Practicum III: Performing and Presenting Scientific Research

**\*Required here if not taken to fulfill Core IMPACTS STEM**

- ASTR 1010 – Astronomy of the Solar System
- ASTR 1010L – Solar System Laboratory
- ASTR 1020 – Stellar & Galactic Astronomy
- ASTR 1020L – Stellar Astronomy Laboratory
- CHEM 1211 – Principles of Chemistry I
- CHEM 1211L – Principles of Chemistry I Laboratory

## **Astrophysics Concentration - 9 hours**

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- ASTR 3061 – Observational Astronomy

**Select two:**

- ASTR 4010 – Stellar Astrophysics
- ASTR 4020 – Astrophysics of Galaxies
- ASTR 4030 – General Relativity & Cosmology
- PHYS 4310 – Particle Physics and the Early Universe

## **Technical or Physics Electives - 6 hours**

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- Six hours of upper-level physics or astronomy courses (up to three credit hours research), computer science, math, engineering, upper-level geography, upper-level astronomy, or upper-level physics courses
- Math courses must be numbered 2000 or above
- PHYS 2901, 2902, and 2903 may count for up to three hours

## **Other Requirements - 6-10 hours**

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- MATH 3000 – Differential Equations
- MATH 3350 – Probability & Statistics
- Modern Language 1002 (if not taken in Core IMPACTS Institution)

## **Electives**

- Electives to bring the total number of credit hours to 120

## Other Requirements

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- [US and Georgia History Requirements](#)
- [US and Georgia Constitution Requirements](#)
- [Regents' Skills Program Requirements](#)

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